

YOUNGJI KOH

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🌐 youngji-koh.github.io

SUMMARY

Researcher specializing in agentic AI, multi-agent reasoning, adaptive human-agent interaction, and multimodal digital health and wellbeing systems.

RESEARCH INTEREST

Agentic Systems • Multi-Agent Reasoning • Human-Agent Interaction • LLM-based Contextual Reasoning • Multimodal Behavioral Modeling • Evaluation & Tooling for Intelligent Agents

EDUCATION

Ph.D. in School of Computing (Expected Aug. 2026)

Sep. 2021 – Present

KAIST, Daejeon, Republic of Korea

Interactive Computing Laboratory (Advisor: Dr. Uichin Lee)

- **Dissertation focus:** Multimodal Behavior Modeling and AI Agent Systems for Mental Healthcare in Intelligent Environments.

M.S. in School of Computing

Sep. 2019 – Aug. 2021

KAIST, Daejeon, Republic of Korea

Software Architecture Laboratory (Advisor: Dr. Sungwon Kang)

B.S. in Computer Science and Engineering

Mar. 2015 – Aug. 2019

Ewha Womans University, Seoul, Republic of Korea

GPA: 4.02/4.30 (*Summa Cum Laude*)

FUNDED RESEARCH PROJECTS

Development of an Early Prediction System for Complex Diseases

Jun. 2025 – Dec. 2025

Institute for ICT Planning & Evaluation (IITP), Lead Researcher

- Designed a comorbidity-aware framework that coordinates disorder-specific agents for diagnosis and intervention planning.
- Focused on integrating LLM-based reasoning, multimodal sensing, and digital health applications for early intervention.
- **Keywords:** LLM Agent · Human-Centered Computing · Multimodal Sensing · Digital Health · Mental Health & Wellbeing

AI Models for Multimodal Mental Health Prediction

Sep. 2023 – Aug. 2025

LG Electronics, Lead Researcher

- Developed deep learning and machine learning models using IoT, mobile, and wearable data for depression and anxiety prediction.
- Evaluated detection accuracy and reliability through large-scale, real-world deployments.
- **Keywords:** Machine Learning · Mental Health & Wellbeing · Multimodal Sensing · Digital Health

IoT, Mobile, and Wearable Data Visualization for Mental Health Self-Reflection

May 2023 – Feb. 2025

LG Electronics, Lead Researcher

- Collected multimodal behavioral and physiological data via IoT, mobile, and wearable devices.
- Designed and evaluated personal informatics dashboards to promote self-reflection and wellbeing.
- The study was featured in national media outlets, including [Yonhap News](#), [Chosun Ilbo](#), [Herald Economy](#), and [Digital Times](#).
- **Keywords:** Data Visualization · IoT & Wearables · Mental Health · Ubiquitous Computing

Robotic Counseling System for Mental Health Support

Feb. 2023 – Oct. 2024

LG Electronics, Research Contributor

- Supported a two-week field study with 20 participants evaluating emotional awareness, engagement, and sense of place in robotic counseling.
- **Keywords:** Social Robots · Mental Health · Human-Robot Interaction · Digital Interventions

Context-Aware Multimodal Smart Speaker for Mental Health Self-Tracking

Sep. 2021 – Sep. 2023

LG Electronics, Lead Researcher

- Designed and developed a context-aware multimodal smart speaker for in-home mental health self-tracking.
- Led in-the-wild deployment for continuous behavioral monitoring and physiological data collection.
- The project received national media coverage highlighting its contribution to digital mental health innovation, including [Yonhap News](#), [Herald Economy](#), [Kookmin Ilbo](#), and [Maeil Business](#).
- **Keywords:** Smart Speaker · Context-Aware Sensing · Smart Home · Digital Health

RESEARCH EXPERIENCE

Undergraduate Research Intern

May 2018 – May 2019

Human-Computer Interaction Laboratory, Ewha Womans University (Advisor: Dr. Uran Oh)

- Build a touchscreen-based mobile application as a prototype focusing on 2D paintings which plays object-level verbal descriptions upon users' touch to enable people with visual impairments to explore and understand various artworks independently
- Design a mobile application for conflict management between lovers on mobile messenger

PUBLICATIONS

Preprints

[J2] A Multimodal Sensor Fusion Approach for Mental Health Detection Using Mobile, Wearable, and IoT Sensors

Youngji Koh, Gyuna Kim, Chanhee Lee, Panyu Zhang, Yunhee Ku, Inhwon Choi, Jewoo Ryu, and Uichin Lee
Under review at *IEEE Journal of Biomedical and Health Informatics (J-BHI)*. IEEE. (Major Revision)

Journal & Conference

[C8] Exploring Data-Driven Approaches to Stress Management: A Systematic Review of Stress Tracking, Intervention, and System Evaluation Methods

Youngji Koh, Gwangyoung Lee, Jeonghyun Kim, Yugyeong Jung, Hwajung Hong, and Uichin Lee
In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)*. CHI'26. ACM. (Accepted)

[C7] Leveraging Smartphone Human Interaction Routine Behavior Task Mining and Modeling for Daily Stress Monitoring

Hansoo Lee, Taehyeon Park, **Youngji Koh**, Jaegil Lee, and Uichin Lee
In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*. UbiComp'26 ACM. (top conference)

[C6] Harnessing Home IoT for Self-tracking Emotional Wellbeing: Behavioral Patterns, Self-Reflection, and Privacy Concerns

Youngji Koh, Chanhee Lee, Eunki Joung, Hyunsoo Lee, and Uichin Lee
In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* (Espoo, Finland, Oct 14-16, 2025). UbiComp'25. ACM. (top conference)

[C5] 'In That Small Space with Just the Two of Us': User Experiences with Cumpa in a Robotic Counseling Center

Chanhee Lee*, Eunki Joung*, **Youngji Koh**, Esther Kim, Sohwi Son, Sunjung Kwon, and Uichin Lee (*: equal contribution)
In *Proceedings of the ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)* (Bergen, Norway, Oct 18-22, 2025). CSCW'25. ACM. (top conference)

[C4] Exploring Context-Aware Mental Health Self-Tracking Using Multimodal Smart Speakers in Home Environments

Jieun Lim*, **Youngji Koh***, Auk Kim, and Uichin Lee (*: equal contribution)

In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)* (Hawaii, USA, May 11-16, 2024), CHI '24. ACM. (top conference)

[C3] Interrupting for Microlearning: Understanding Perceptions and Interruptibility of Proactive Conversational Microlearning Services

Minyeong Kim*, Jiwook Lee*, **Youngji Koh**, Chanhee Lee, Uichin Lee, and Auk Kim (*: equal contribution)

In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)* (Hawaii, USA, May 11-16, 2024), CHI '24. ACM. (top conference)

[J1] Deep Learning-based Bug Report Summarization Using Sentence Significance Factors

Youngji Koh, Sungwon Kang, and Seonah Lee

Applied Sciences 2022

[C2] Bug Report Summarization using Believability Score and Text Ranking

Youngji Koh, Sungwon Kang, and Seonah Lee

International Conference on Artificial Intelligence in Information and Communication (ICAIIIC 2021). IEEE.

[C1] Deep learning based bug report summarization using sentence assessment score

Youngji Koh and Sungwon Kang

Korea Software Congress (KSC 2020) *Best Paper Award*

Poster & Workshop

[W1] Data Visualization for Mental Health Monitoring in Smart Home Environment: A Case Study

Youngji Koh, Chanhee Lee, Yunhee Ku, and Uichin Lee

IEEE VIS 2023 Workshop on Visual Analytics in Healthcare

[P4] LV-Linker: Supporting Fine-grained User Interaction Analyses by Linking Smartphone Log and Recorded Video Data

Hansoo Lee, Sangwook Lee, **Youngji Koh**, and Uichin Lee

ACM Symposium on User Interface Software and Technology (UIST 2022)

[P3] Supporting Object-level Exploration of Artworks by Touch for People with Visual Impairments

Nahyun Kwon, **Youngji Koh**, and Uran Oh

ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2019)

[P2] Analysis of Conflicts in Romantic Relationships on Mobile Messengers

Young-Ji Koh, Hyunggu Jung, and Uran Oh

HCI Korea 2019

[P1] Real-time Fire Evacuation System based on Indoor Location Tracking and Route Optimization

Young-Ji Koh, Naeun Go, ChaeYoon Kim, SooHee Lee, and Sangsoo Park

Korea Software Congress (KSC 2018) *Student Paper Competition, Merit Award*

HONORS AND AWARDS

Public Relations Award

[Donga Science](#), Korea

Dec 2025

Creative Research Award

K-Data Science Conference, Korea

Sep 2025

Special Recognition for Outstanding Reviews

ACM CHI

Dec 2024

Ewha Womans University Scholarship
Merit-based scholarship

Spring 2018

TEACHING EXPERIENCE

Teaching Assistant, KAIST
CS492 Data Visualization

Fall 2021, Spring 2023

Teaching Assistant, KAIST
CS204 Discrete Mathematics

Spring 2020

COMMUNITY SERVICE

HCI@KAIST
Student Steering Committee

2024, 2025

ACADEMIC SERVICE

Reviewer

ACM Transactions on Computing for Healthcare
IEEE Transactions on Computational Social Systems
ACM UbiComp/ISWC Notes and Briefs
ACM CHI